

Internal and External Determinants of Earnings Management: A Corporate Governance Approach for Emerging Markets

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Abstract

Under a corporate governance approach this paper analyzes the internal and external determinants on the earnings management for a sample of Latin American companies over the years 1997 to 2013. Since we account with cross-sectional and time series information, we used panel data for the empirical analysis to deal properly with the endogeneity problems. Our analysis reveals that the capital structure, the dividend policy and the ownership structure impact companies' earnings management. We found an unexpected result in relation with the external corporate governance mechanisms. Based on our findings we can conclude that internal and external corporate governance systems have a significant influence in the capacity of managers to manipulate earnings.

JEL Classification: G30, G32, G34

Key words: Earnings management; Discretionary accruals; Corporate governance; Emerging markets; Legal system; Financial development

1.INTRODUCTION

We are not the first in studying the impact of corporate governance systems on the managerial performance, and particularly on the earning management.¹ The lack of monitoring efficiently the managerial performance, the eradication of opportunistic behavior, and the application of fair and right –not perverse– incentives is a real need in nowadays more complex capital markets. A tangible example of this has been the lack of efficient corporate governance systems which has caused the recent financial crisis with negative consequences worldwide. Accounting scandals like Enron, Worldcom, Parmalat, Lehman Brothers and others, revealed the necessity of reinforcing the rules and regulations towards a more transparent disclosure of the financial statements. In a few words, the financial markets could not anticipate the consequences of these scandals because the accounts were

¹ There are two kinds of earning management techniques: the accounting methods and the operating methods (Tirole, 2006). This work is only focused on the accounting methods.

opportunistically manipulated. The main goal of this paper is to analyze under a corporate governance approach the impact of the internal and external determinants on the managers' discretionary behavior. Specifically talking, on the one hand, at the firm level we study how the capital structure and the dividend decisions, as well as the ownership structure determine the accounting discretion; whilst on the other hand, at the country level variables we included the developed of the financial system as well as the legal and regulatory systems as determinants of the managers' discretionary behavior.

The vast majority of the empirical studies are based in institutional environments of developed countries and there is not much research for emerging countries. Therefore we want to contribute to the literature with an analysis of emerging countries such as Argentina, Brazil, Chile, Colombia, Mexico, and Peru over the years 1997 to 2013. Among the major findings, we found evidence the firms manipulate discretionally their results; either by increasing profits or reducing them. We found also that the firm level variables are important drivers of the earnings management. However, unexpectedly, we found a positive relationship between the level of financial development and the earnings management. This is due to the fact that in Latin America financial markets are far from being efficient and managers take advantage of it to manipulate earnings in any direction.

This study contributes to the literature in the following ways. First, it extends the empirical literature focusing the analysis on emerging countries. Second, it does consider in the analysis not only internal or firm-level determinants, but also external or country-level determinants of the discretionary accruals that very few papers deal with. Third, it uses a novel statistical technique which allows efficiently dealing with problems of endogeneity and unobserved heterogeneity of firms included in the sample. Particularly, we use panel data which are a mix of cross sectional and time series data. Forth, this study establishes a series of recommendations for policymakers and practitioners. Unfortunately, the unavailability of information has not allowed us to analyze board of directors as an additional –and not less important– internal corporate governance system.

The reminder of the paper includes in the second section the theoretical framework and research hypotheses. The third section displays the methodology. Section four describes the results and in the last section we present the most relevant conclusions. The appendix describes the variables construction.

2.THEORETICAL BACKGROUND

a. Agency conflicts and earnings management

Agency approach is the theoretical body which studies the contractual conflicts caused by the different incentives between the contractual parties (M. Jensen, 1994; M. C. Jensen & Meckling, 1976). In its more fundamental idea, these problems are basically asymmetries of interest between the parties. Jensen and Meckling (1976) build up their theory of the firm based on the agency costs. These agency costs are the monitoring costs incurred by shareholders because their managers are prone to run the company in their own interest instead of the maximization of the wealth of current shareholders. This conflict of interests demands for mechanisms which encourage managers to behave in the best interest for shareholders instead of the managers' interests. This set of mechanisms is named corporate governance systems.

Broadly defined, corporate governance corresponds to the set of rules in capital markets governing the equity investments in firms (Claessens & Yurtoglu, 2013). This would include listing requirements, insider dealing arrangements, disclosure of information and accounting rules, and protections of (minority) shareholder rights. All these rules are basically settled by external mechanisms which are away from the firm's control. Nevertheless, the effect of the corporate governance trespasses into the firm in the form of internal mechanisms which are a matter of control at the firm-level. In other words, insiders might use efficiently these tools in order to realign the interests of managers with those of shareholders. This conceptualization of corporate governance is close to the one advised by Shleifer and Vishny (1997) as the way used by suppliers of finance to assure the return on their investment. Similarly, the interpretation of the Zingales (1998)'s approach on corporate governance is defined as the complex set of constraints –contracts– that shape the ex-post bargaining over the quasi rents –cash flow rights– generated by the firm. Since the goal of the paper is the analysis of internal (firm-level) and external (country-level) determinants, we are going to use a wide approach of corporate governance, therefore.

As corporate governance systems are, by default, imperfect; managers have incentives to make decisions discretionarily by following their own interests instead of following the wealth maximization of their shareholders. One of the mechanisms used to convey information to potential investors and capital markets is the financial reporting.

Therefore, a plausible reaction of managers is the choice of techniques to manage the earnings (M. Jensen, 2003; Smith, 1976). According to Shen and Chih (2007) earning management is the alteration of firms' reported economic information and performance to either mislead stakeholders in order to reduce outsider interference and protect insiders' private control benefits. The techniques to manage the earnings might take the form of changes in the accounting methods (Moses, 1987) such as modifications in the inventory valuation system or in accounting methods

recording and recognizing extraordinary income and expenses (Beattie et al., 1994); the smoothing of earnings in real activities such as in import relief investigations (J. J. Jones, 1991) or in seasoned equity offerings (Cohen & Zarowin, 2010); in the choice of reserves or provisions for loan losses which are always subjective (Tirole, 2006); or in the accruals (Dechow, 1994; Gabrielsen, Gramlich, & Plenborg, 2002; García-Teruel, Martínez-Solano, & Sánchez-Ballesta, 2009); among many others. Out of all these, perhaps the one which has raised more attention in the empirical literature are the accounting accruals (Dechow, 1994; Dechow, Sloan, & Sweeney, 1995; J. J. Jones, 1991; S. Jones & Sharma, 2001). The main goal of accounting accruals is to improve the quality of the information reported in the financial statements and to avoid the mismatching between cash flows and the flow of income and expenses. Nevertheless, there might be a discretionary use of the managerial capacity to manipulate or redirect the information reported in the financial statements (M. E. Barth, Cram, & Nelson, 2001), named discretionary accruals.

The discretionary accruals are perhaps the most common way to manage the earnings. The main motivations for the opportunistic earnings management are, among others, first, the compensation schemes (Faulkender, Kadyrzhanova, Prabhala, & Senbet, 2010). Compensation of executives is embedded to the firm performance. Due managers have inside information; they have opportunities to manage net income to maximize their bonuses. Consequently, it is more likely that managers will incur in current-period earnings management to ensure better future compensation packages. Second, the constraint imposed by debt covenants. If the firm violates the covenants, it might face higher cost of debt and reject potentially valuable investment projects –those with positive net present values, for instance– (DeAngelo, DeAngelo, & Skinner, 2004). Third, managers might overstate the earnings as a result of beating or achieving the analysts' forecasts. Barth and Hutton (2004) find that stock prices of firms with higher analyst following more rapidly incorporate information on accruals and cash-flows than prices of less followed firms. In the same way, the mere fact that the analysts are tracking the firm's performance puts further pressure on firms to maintain earnings momentum to fulfill the market expectations (M. E. Barth, Elliott, & Finn, 1999). Fourth, managers might also manage earnings to set better prices before IPOs, or to reduce the negative impact of loan losses; among other motives.

Whereas non-discretionary (or normal) accruals are basically means to improve the content of the accounting information; the discretionary (or abnormal) accruals are means to manipulate intentionally earnings in favor of managers' interests. Therefore, the financial reports may be meaningless if accounts are seriously manipulated or misrepresented.

Since the origin of discretionary accruals is based on the ability and incentives of managers to manipulate the financial statements, on the one hand, and on the efficiency (or lack of it) of the corporate governance mechanisms, on the other hand; the goal of this paper is to study how the variables associated to corporate governance systems impact on the discretionary use of accruals to manage the earnings.

b. Determinants of earnings management

i. Firm-level determinants

At the firm level three determinants will be studied: i) capital structure decisions; ii) dividend policy decisions; iii) ownership structure

Capital structure decisions: The capital structure decisions might be a double edged sword on earnings management. On the one hand, the debt level might be used as a control mechanism of managerial behavior. The preference of managers for the consumption of perks at the expense of the shareholders' wealth is alleviated through more leveraged capital structures. Higher levels of debt reduce the opportunistic managers' power by reducing the cash flow available for spending in discretionary worthless assets, such as perks or negative net present value projects (Frank & Goyal, 2009; Harris & Raviv, 1991; M. C. Jensen, 1986). For Hunsaker (1999) an increase in leverage increases also the bankruptcy risk which reduces the managerial consumption of resources in activities which do not add value. Consequently, managers will avoid this opportunistic behavior in favor of maximizing the interests of their current shareholders. Therefore, one might expect that firms with more leveraged capital structures should observe less earnings management in the form of discretionary accruals.

Nevertheless, the capital structure decisions might play against the interest of shareholders and in favor of the managers' opportunistic behavior. Managers might cause a good impression through the financial statements in order to issue debt in more favorable conditions and/or achieve the goals imposed by the debt covenants (Mohrman, 1996). Consequently, firms closer to violating debt covenants manage earnings more aggressively (Shen & Chih, 2007). Considering these arguments, the fostering or constraining roles of debt on earnings management –named as leverage effect and the reverse leverage effect by Shen and Chih (2007)– seem to be an empirical issue and both a positive and a negative effect can be justified.

H1: It is expected either a positive or negative relationship between the leverage and the earnings management.

Dividend Policy Decisions: The decision of how much to pay out as dividends to current shareholders and what proportion of earnings should be reinvested in the form of retained earnings is one of the most important decisions faced by

managers. The seminal work of Lintner (1956) documents that managers are reluctant to cut dividends and target long-term pay-out ratios (defined as dividend conservatism) when making dividend decisions. Similarly, using a survey sample of 256 public companies and 128 private firms, Brav et al. (2005) investigate payout policies in the 21st century. Their analysis indicates that, consistent with dividend conservatism, most of the dividend-payers have a strong desire to avoid dividend reductions and to smooth dividend streams from year to year. They suggest that maintaining dividend levels is the main variable in deciding dividend policies; while pay-out ratios are of secondary importance. They also conclude that since dividend reductions have negative consequences, it is plausible that managers would manipulate the accounting information in order to keep smooth the dollar dividends. Recently, Anglin et al. (2013) found evidence that dividend policy impacts both upward and downward real earnings management. In addition, they stated that payers manipulate earnings upward through real activities to mitigate the shortfall of pre-managed earnings relative to prior year dividends when pre-managed earnings are lower than dividends paid in the prior year, suggesting that dividend levels are an important earnings benchmark. They also documented a stronger relationship between changes in pre-managed earnings and real earnings management for payers than for non-payers, suggesting that dividend policies impact real earnings management. Consistent with the importance of dividend policy in real earnings management, they showed that dividend payers that follow conservative dividend policies manipulate earnings to a greater extent than dividend payers that do not follow conservative dividend policies. Therefore, according to the previous arguments:

H2a: It might be expected either a positive or a negative relationship between the discretionary accruals and the cash dividends.

The evidence on the interaction between dividend policy and governance structures comes also from cross-countries analyses where several studies document that the institutional and legal environments affect firms' payout policies. For instance, La Porta et al. (2000) report that firms in common-law countries pay higher dividends than those operating in civil-law countries where minority shareholders suffer from weaker legal protection. In fact, according to Mitton (2004), the preference for dividends may be stronger in emerging markets with weak investor protection where shareholders perceive a greater risk of expropriation by insiders in such countries. In the same line, (although for a sample of firms of the developed market of Japan) He (2012) finds out that the dividend pay-out policy depends substantially on the capacity of the country's legal system to constrain both controlling shareholders and insiders to pay dividends to minority claimants (shareholders). Thus, we cannot dissociate the nature of the dividend policy and the

characteristics of the legal system across countries. Consequently, our next hypothesis states that:

H2b: In institutional contexts with weak corporate governance systems it might be expected more earnings management to achieve certain cash dividend goals than in institutional contexts with relatively stronger governance systems.

Corporate ownership structure: The ownership structure plays a critical role in the firm's governance. In fact, Bebchuk and Hamdani (2009) argue that all the effort invested in creating a composite index of corporate governance is worthless unless ownership structure is not incorporated. Consequently, corporate ownership structure cannot be dissociated from corporate governance nor from its impact on the opportunistic behavior of managers.

When owners are also managers of the firms there is an overlap between ownership and control and consequently the potential agency problems are minimized. Warfield et al. (1995) show that insiders' ownership has a significant effect on managers discretionary decisions. They show that managerial ownership is positively associated with earnings' explanatory power for returns and inversely related with the magnitude of accounting accruals adjustments. Consistent with Morck et al. (1988), in the lowest levels of insiders' ownership there is an alignment of interests which means a negative relation between abnormal accruals and insiders' ownership. Nevertheless, an entrenchment effect is found for the highest levels of insiders' ownership. Similar results are found by Ali et al. (2008) for Asian firms, who show that managerial ownership is negatively associated with the magnitude of accounting accruals. Consequently, when managerial ownership is considered, a non-monotonic relationship might exist between this and the earning managements.

H3a: At low levels of insiders' (managerial) ownership, alignment of interests exists and the likelihood of earning managements is lower. Nevertheless, at high levels of insiders' ownership the risk of entrenchment increases and consequently the likelihood of managing opportunistically the accounting earnings. This means that a U-shaped relation is expected between insider's ownership and earnings management.

It has been widely argued that concentrated ownership structures solve some agency problems through direct supervision of managers (Ang, Cole, & Lin, 2000; Brown, Beekes, & Verhoeven, 2011). Dispersion produces free-rider problems and wrong incentives on minority shareholders for monitoring (Bottazzi, Da Rin, & Hellmann, 2009). This argument leads to a positive relationship between ownership concentration and firm value as posted by the monitoring hypothesis. Consequently, the earnings management behavior is reduced as a result of more concentrated ownership structures. Therefore, it can be derived that the vertical

agency conflict between managers and shareholders, and the consequent management of accounting earnings, might be efficiently reduced through a higher ownership concentration (M. C. Jensen & Meckling, 1976; Shleifer & Vishny, 1986). Empirically, López and Saona (2005) for Chilean firms, and the recent work of Alves (2012) for Portuguese firms, they both find that discretionary accruals as a proxy for earnings management are negatively related to ownership concentration. Therefore, our following hypothesis states that:

H3b: The higher the ownership concentration the lower the room for opportunistic behavior through earnings management.

ii. Country-level determinants

Accounting standards: The adaptation of the accounting systems to international standards is not new. For instance, in general, the extant literature has found a positive impact of the IFRS adoption on accounting quality. Positive accounting research provides evidence that the accounting policy choices made by firms are determined not only by the regulatory systems but also by factors that are specific to the firm, including its operating circumstances and managerial preferences, all of which will result in a diversity of accounting treatments (Beattie et al., 1994). Accounting rules can limit a manager's ability to distort reported earnings; but the extent to which rules influence the reported earnings depends always on how well the rules are enforced (Leuz, Nanda, & Wysocki, 2003). The recent work of Palacios and Martínez (2014) examines whether the adaptation of Mexican firms to IFRS standards impacts the managers capacity to manage discretionally the earnings. Their results, however, suggest that the convergence to international accounting standards have increased the comparability of accounting earnings; but such adoption of international accounting standards is not associated with lower levels of discretionary accruals. This suggests that the choice of the accounting method could be driven by the opportunistic behavior of managers. What it is expected from the accounting standards is that they reported information in the financial statements be more comparable and clean of potential discretionary adjustments. Empirically, Barth et al. (2008) find that firms applying International Accounting Standards from developing and developed countries generally evidence less earnings management, more timely loss recognition, and more value relevance of accounting amounts than do matched sample firms applying non-U.S. domestic standards. Bilgic and Ibis (2013) find that after new reporting standards, there is an increase in the value relevance of earnings and book values together for Turkish firms, and this increase is mainly due to the increase in the value relevance of book values after the application of the IFRS. In the same way, for a sample of Asian firms, Shen and Chih (2007) find that wealthy countries have less incentives to

adopt earnings smoothing, representing the effect of better accounting/reporting standards in more developed countries. Therefore, it is presumed that companies using international accounting standards to have greater incentives to report transparently because they are subject to tighter restrictions and exposed to a higher litigation risk. Consequently, the hypothesis concerning the adoption accounting standards is:

H4: Less discretionary earning management should be observed under the adoption of international accounting standards.

Legal system: The protection of outsiders depends on the conjunction of the existence of a legal body and the enforcement of the law. Leuz et al. (2003) provide evidence that countries with lower investor protection usually have a higher magnitude of earnings management and this lack of efficient protection gives insiders more incentives to obfuscate firm performance. Ball et al. (2003) argue that the institutional arrangements of a country are the most important factor in controlling managers' self-interest, reducing opportunistic behavior translated into earning management, and improving the quality of the accounting information. Sarbanes-Oxley Act of 2002 (SOX) in the United States is a clear example of these regulatory arrangements.² The major aim of the SOX legislation is to limit corporate fraud, especially fraud associated with manipulation of financial statements and insider trading. Particularly, SOX imposes increased penalties on corporate executives charged with manipulation of financial statements and fraudulent insider trading (Chhaochharia & Grinstein, 2007). Therefore, one might expect that the corporate governance rules limit the corporate executives' discretionary behavior in managing earnings. In fact, in an event study Chhaochharia and Grinstein (2007) find that firms that were less compliant with the rules realize a greater value improvement compared to firms that are more compliant with the rules. Cohen et al. (2008) provide evidence that accrual-based earnings management increased steadily until before the passage of the SOX Act, and that this increases were concurrent with increases in equity-based compensation.

In the same way, Ball et al. (2000) analyze the effect of the institutional legal systems such as the comparison between the common-law and the code-law on the quality of accounting earnings. They find that common-law accounting income does indeed exhibit significantly greater timeliness than code-law accounting income, but that this is due entirely to greater sensitivity to economic losses (income conservatism). Consequently, we can derive out of this that the legal origin of the country might determine also the way the executives use their discretionary capacity to manage the accounting earnings.

² This regulation came after a series of financial scandals involving accounting irregularities and share price manipulation in several leading companies that time, such as the energy company Enron, Worldcom, and Tyco, among many others, which filed restated financial results with the Securities Exchange Commission (SEC) during 2001 and 2002.

As it was mentioned before, not just the legal system is relevant in driving earning management, but also the enforcement of that legal body. Demirgüç-Kunt and Maksimovic (1998) and later on Demirgüç-Kunt et al. (2004) find that better legal enforcement and efficient regulatory systems are associated with lower levels of corruption which make the financial system to perform with much less frictions. Although focused on financial institutions only, Naceur and Omran (2011) study the influence of bank regulation and concentration on the value of Middle East and North Africa commercial banks. They find that regulatory and institutional variables seem to have an impact on bank performance. In the same way, Shen and Chih (2007) suggest that a firm in a country with good anti-director rights does less earnings smoothing. This allows us to suggest the following hypotheses:

H5: It is expected a negative association between both the legal enforcement and the efficiency of the regulatory system and the extent of earnings management.

Financial development: The role of financial institutions in financial markets is to serve as a middleman between saving and borrowing units by reducing the transaction costs. The financial development enhances the allocation of capital, liquidity, the access of firms to more sophisticated financial instruments, the flows of information, and reduces the cost to external financing, better enabling firms to exploit current growth opportunities (Love, 2011). It has been widely accepted in the literature the influence of the development of a country's financial sector on the level and growth rate of its per capita income (Rajan & Zingales, 1998) and in the aggregate real economic activities such as GDP growth, and the growth rate of productivity and investment at country-level (De Nicolò, Laeven, & Ueda, 2008); in the evolution of accounting regulation (Rajan & Zingales, 1998); the availability and allocation of financing for growth opportunities (Gonenc & de Haan, 2014; Kirch & Soares, 2012); and the extent of earnings management (Enomoto, Kimura, & Yamaguchi, 2014), among others.

Concerning the discretionary behavior of corporate executives to manage earnings, it is likely that financial development enhances the monitoring and scrutiny of accounting figures by the market participants, because of strengthened laws and regulations for investor protection and by extension, due more sophisticated market participants. Empirically, for a sample of 37 countries, Enomoto et al. (2014) show that managers are restrained with regards to earnings management under higher levels of financial development. They support this relationship on three arguments. First, they argue that higher quality accounting information is needed in countries with more developed financial systems; second, financial development disciplines managers and mitigates their incentives to manage earnings; and finally they say that there is a correlation between financial development and accounting institutions in each country.

When financial markets are not well developed, market anomalies and opportunistic behavior arises, such as the discretionary management of accounting earnings. For instance, the recent work of Degeorge et al. (2013) shed more light on the role of financial system on the analyst coverage and earnings management. For a sample of more than twenty countries they provide evidence that increased analyst coverage is associated with less earnings management in countries with high levels of financial development; while in less financially developed countries, the analyst coverage is not associated with reductions in earnings management.

Consequently, it seems to be that the deepness of the financial development impacts asymmetrically the extent of earnings managements. Therefore, we might hypothesize that:

H6: It is expected that the earnings management be lower before more developed financial systems.

3.METHODOLOGYAND VARIABLES MEASURE

a. Methodology

Since we account with a data set of cross-sectional and time series information, we chose to use panel data for the empirical analysis. The statistical analysis is developed with a sample of 712 firms from Argentina, Brazil, Chile, Colombia, Mexico, and Peru. The composition of the panel data is described in Table 1. The panel data includes 4,550 observations over the years 1997 to 2013 with an average of 6.39 observations per firm. Recall that a minimum of 4 observations per firm is required to use efficiently the information in panel data. The microeconomic data at firm level corresponds basically to the financial reports and market information (e.g. price per share, stock market capitalization, outstanding stocks, among others) gathered from Thomson Reuters dataset. The macroeconomic information at country level was obtained from the updated data based on the work of Beck et al. (2000) publicly available at the World Bank web page. This dataset reports basically information about financial development by country and year.³ Worldwide governance indicators concerning the accounting standards and the legal system by country was obtained from the updated work of Kaufmann et al. (2011) whose data set is also publicly available.⁴

Due to the panel structure of our data, which is a combination of cross sectional and time series information, we have used in the econometric analysis the generalized method of moments (GMM). The panel data methodology allows us to control for two basic problems in this kind of studies: the unobservable

³ The latest update took place in November 2013. Information can be downloaded from the permanent URL <http://go.worldbank.org/X23UD9QUX0>

⁴ The latest update took place in September 2014. Information can be download from www.govindicators.org

heterogeneity and the endogeneity problems (Manuel Arellano, 2002; Manuel Arellano & Bover, 1990). The relationships between the firms' characteristics and their impact on the earnings management must be interpreted carefully because of the possibility of observing spurious relations which foster endogeneity problems.⁵ These models might also suffer from the unobserved heterogeneity problem, where the identified relationships are symptoms of some unobservable factor that drives both the dependent and independent variables (e.g. the managerial style, firm culture).

Because in both of these problems the independent variables are endogenous and correlated with the residuals of the regressions, the OLS estimation is both biased and inconsistent (Brown et al., 2011). As a result, we address the endogeneity and unobservable heterogeneity problems in the estimations by using the GMM estimator proposed by Blundell and Bond (1998) and Bond (2002) which might provide further efficiency gains. This approach works through two simultaneous equations involved in system estimator. The first equation considers all the information in levels and the second equation is in first differences where the instruments are the lags of the endogenous variables (Baum, Schaffer, & Stillman, 2003; Bond, 2002; Wooldridge, 2002).

Several statistical tests are used. The Hansen/Sargan test assesses the model specification validity (Hansen, Heaton, & Yaron, 1996). This test examines the lack of correlation between the instruments and the error term. The AR1 and AR2 statistics measure the first- and second-serial correlation, respectively. The Wald test of joint significance is also used to assess the significance of all the independent variables in the sample. The Lind-Mehlum contrast is used to study the statistical significance of the non-monotonic relationships suggested in this study (e.g. the particular case of the corporate ownership structure).

b. Earnings management measures and independent variables

Two alternative estimations of the earnings management are used. For the measure of opportunistic earnings management we use the absolute discretionary accruals. Since total accruals are known, the discretionary accruals must be estimated (see Models 1 and 2 below).

i. Model 1:

⁵ An exogenous variable is that whose values are given and are not affected by the variable to be explained, which is said to be endogenous. As a result, there is an endogeneity problem when some of the explanatory variables are not strictly exogenous.

Based on Dechow et al. (1995) the total accrual (ACC_{it}^{Mod1}) (with the super index Mod1) denotes the component of earnings for each i firm during the t period is computed as:

$$ACC_{it}^{Mod1} = (\Delta CA_{it} - \Delta Cash_{it}) - (\Delta CL_{it} - \Delta STD_{it}) - Dep_{it} \quad (1)$$

Where CA denotes the current assets, $Cash$ is the cash and cash equivalent, CL are current liabilities, STD stands for the short-term debt and the current proportion of long-term debt, and Dep is the annual depreciation expense.

Once the total accruals are calculated, we have to split them into their non-discretionary and discretionary components. Non-discretionary accruals are aimed to improve the informational content of financial statements, so we could wonder about the factors that cause these normal adjustments. According to the Jones (1991)'s model, total accruals are affected by the firm's usual business (which can affect non-cash current assets and liabilities) and by fixed assets (which can affect the depreciation expense). Consequently, ACC are regressed depending on the change in sales ($\Delta Sales_{it}$) and the gross level of property, plant, and equipment (PPE_{it}) in the following equation:

$$\frac{ACC_{it}^{Mod1}}{A_{it-1}} = \beta_0 + \beta_1 \frac{\Delta Sales_{it}}{A_{it-1}} + \beta_2 \frac{PPE_{it}}{A_{it-1}} + \varepsilon_{it} \quad (2)$$

Regarding the expected signs for β_1 and β_2 it might be said that this is not trivial, except for β_2 . In this case, it is expected that β_2 be negative because depreciation has been included with negative sign in the definition of total accruals (ACC). However, there is no a clear prediction for the sign of β_1 because, on the one hand, higher level of sales might imply higher accounts receivable but, on the other hand, increases in sales usually imply increases in short-term debt too, so the net effect on the working capital might not be determined a priori.

So the value of (ACC) in equation (2) is the level of total accruals depending on the firm's activity and the composition of the firm's assets. Therefore, the error term in the regression, which is the difference between observed and estimated accruals as stated in equation (3), would become the part of total accruals due to the discretionary behavior of managers. So the discretionary accruals ($DACC_{it}^{Mod1}$) should take the form:

$$\left| \frac{DACC_{it}^{Mod1}}{A_{it-1}} \right| = \frac{ACC_{it}^{Mod1}}{A_{it-1}} - \left(\hat{\beta}_0 \frac{1}{A_{it-1}} + \hat{\beta}_1 \frac{\Delta Sales_{it}}{A_{it-1}} + \hat{\beta}_2 \frac{PPE_{it}}{A_{it-1}} \right) \quad (3)$$

Where $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ are the estimators for β_0 , β_1 , and β_2 coefficients, respectively. Since the discretionary behavior in earning management might be used either to increase or reduce the earnings, we follow Gabrielsen et al. (2002) and

calculate the absolute value for $DACC$ to measure the extent of this discretionary behavior instead of its direction.

ii. Model 2:

In the same way as before, this is also a cross-sectional model of discretionary accruals based on Jones (1991)'s model as described in Dechow et al. (1995):

$$\frac{ACC_{it}^{Mod2}}{A_{it-1}} = \beta_0 + \beta_1 \frac{\Delta Sales_{it}}{A_{it-1}} + \beta_2 \frac{PPE_{it}}{A_{it-1}} + \varepsilon_{it} \quad (4)$$

Where ACC_{it}^{Mod2} denotes the total accruals defined as the difference between earnings before extraordinary items and discontinued operations and operating cash flows taken from the cash flow statement. The other variables are the same as those in equation (2).

The coefficient estimates from equation (4) are used to estimate the firm-specific non-discretionary accruals as:

$$NDACC_{it} = \hat{\beta}_0 \frac{1}{A_{it-1}} + \hat{\beta}_1 \frac{\Delta Sales_{it} - \Delta AR_{it}}{A_{it-1}} + \hat{\beta}_2 \frac{PPE_{it}}{A_{it-1}} \quad (5)$$

Where ΔAR_{it} is the change in accounts receivable from the preceding year. Following Cohen et al. (2008), while computing the non-discretionary accruals, we adjust the reported revenues on the sample firms for the change in accounts receivable to capture any potential accounting discretion arising from sale credits. Then the measure of discretionary accruals, in the same way as in Model 1, is the difference between total accruals and the fitted non-discretionary accruals ($DACC_{it}^{Mod2}$), defined as:

$$\left| \frac{DACC_{it}^{Mod2}}{A_{it-1}} \right| = \frac{ACC_{it}^{Mod2}}{A_{it-1}} - \left(\hat{\beta}_0 \frac{1}{A_{it-1}} + \hat{\beta}_1 \frac{\Delta Sales_{it} - \Delta AR_{it}}{A_{it-1}} + \hat{\beta}_2 \frac{PPE_{it}}{A_{it-1}} \right) \quad (6)$$

iii. Independent variables

Among the independent variables we have those directly linked to our research hypotheses such as the capital structure, the dividend policy, and the ownership structure in addition to the IFRS, the development of the financial system and the legal and regulatory systems. Control variables were introduced in the model in order to lessen the subidentification problems. The control variables are the firm size, profitability, default risk, industry dummy variables, and time and country dummy variables. The last two variables are needed because of the panel data structure of the econometric models. Details about the calculation of the variables are displayed in the Appendix.

We use two alternative measures for the firm's capital structure: the leverage at book value (LevB) and the leverage at market value (LevM). Nevertheless Bowman's (1980) study provides empirical evidence on the comparability of the book-value and market-value measures of leverage in association tests on systematic risk; Parsons and Titman (2008) argue that the use of leverage at book value or market value is an issue of debate. In fact, Bowman (1980) suggests that accounting (book) measures of debt are statistically indistinguishable from market-value measures. In the same line, Welch (2011) points that the findings of most papers are likely to hold regardless of the leverage definition. However, according to Lang et al. (1996), a measure based on market values could give too much importance to the recent changes in the equity. So, since the debate on the use of book versus market measures of leverage is clearly on the table, we prefer to use the two of them as robustness measures.

The variables used for the dividend policy correspond to two alternative measures of the payout ratio. The first one (Div1) is computed based on the dividends per share and the earnings per share and the second one (Div2) is calculated according to Pindado and De La Torre (2006)

For the ownership structure we use also two alternative measures for robustness purposes. The first one includes the ownership closely held (InsOwn) which is represented by the shares held by insiders. The second corresponds to the corporate ownership concentration (Own1) represented by the proportion of shares in hands of the majority shareholder.

Among the control variables we included the firm size (Size), the profitability (ROA), and the default risk. For the default risk we opted for two alternative measures (Z1 and Z2). The first one (Z1) is the ordinary Altman (1968)'s Z-Score which was originally developed for US manufacturing companies. Since our sample is compounded by Latin American firms, we decided to use also the version of the score (Z2) for emerging markets developed later on (Altman, 2005).

At the country-level determinants we have the following institutional variables: accounting standards, the legal system and the financial development.

For the accounting standards we use a dummy variable (IFRS) which takes the value 1 if the firm reports its financial statements under the International Financial Report System and 0 otherwise.

Following Kaufmann et al. (2011), for the legal system we include the following variables, resulting in a total of six dimensions of governance: (i) Voice and Accountability (VA) which is the process by which governments are selected, monitored, and replaced; (ii) Political Stability and Absence of Violence/Terrorism (PS) which measures the perceptions of the likelihood that the government will be

destabilized or overthrown by unconstitutional or violent means, including politically-motivated violence and terrorism; (iii) the Government Effectiveness (GE) corresponds to the quality of public and civil services, and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies; (iv) Regulatory Quality (RQ) which measures the perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development; (v) Rule of Law (RL) which reflects the confidence of the agents in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence; and finally (vi) the Control of Corruption (CC) which measures the perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as "capture" of the state by elites and private interests. Although these original six indicators range from approximately -2.5 (weak) to 2.5 (strong) governance performance, in our sample the values are not that extreme.

We use several measures taken from the updated dataset of Beck et al. (2000) to analyze the level of financial development. For instance, we used six indicators for the development of the financial system (three to measure the extent to which the banking sector is developed, and another three for the development of the capital markets): i) Central bank assets to GDP (CBAGDP); ii) Other financial institutions assets to GDP (OIAGDP); Private credit by deposit money banks to GDP (PCGDP); Stock market capitalization to GDP (SMKGDP); International debt issues to GDP (IDGDP); and Private bond market capitalization to GDP (PBGDP). For all these six indicators, it is expected a more developed financial system as the indicator increases.

4.RESULTS

a. Descriptive statistics

Table 1 describes the composition of the panel data used in the empirical analysis. Table 2 displays the alternative measures of the discretionary accruals and the independent variables. The table shows the number of observations and the mean as per variable. We can observe that the mean value for the discretionary accruals are always higher than 0. In table number 3 we have tested the hypothesis whether such values for the discretionary accruals are statistically different from zero. We can observe in the panel A that such mean values are in fact statistically significant from zero. In the same way to García-Meca & Sánchez-Ballesta (2009), this particular finding provides evidence that in the countries sample the firms manipulate discretionally their results; either by increasing profits or reducing

them. The other panels in the table 3 describe the mean values for our two alternative measures for the discretionary accruals by country. Similarly, in all the cases we observe that by country, firms do manipulate their earnings.

Table 2 displays that the leverage position (LevB) is about 48.65% of their total assets. At market values, this ratio (LevM) is 17.2%. The market value of this ratio is comparatively lower than the book value due to the market capitalization of Latin American firms is overvalued (Lefort & Walker, 2007; P. Saona & San Martín, 2015). In addition to that, we observe that firms do payout a large proportion of their earnings in the form of dividends (Div1) in emerging markets. This is in line with the argument that in emerging markets with weak protection of the investor rights, shareholders demand for large dividend yields and consequently firms pay for large cash dividends. This can be seen as a short term approach of investment from the shareholders' point of view, which is looking for a short payback period in order to recover their investment soon. This weak protection to the shareholders rights is also reflected on the corporate ownership structure (Espinosa, 2009; Khanna & Palepu, 2000; Lefort, 2005), which is characterized by the presence of large controlling shareholders, pyramidal structures, institutional investors (e.g. Mutual funds), and highly concentrated corporate ownership. In fact, the descriptive statistics shows that the shares in hands of the controlling shareholder and in hands of the executives (InsOwn) represent 58.71% of the outstanding shares; whilst the majority shareholder has about 27.13% of the ownership (Own1).

Table 4 provides the correlation matrix results for the earnings management measures and the independent variables. For the leverage at book value we observe a negative correlation with the earnings management. The payout ratio, as well as the concentration of the ownership, is both negatively correlated with the earnings management measures. For most of the country-level determinants –those that measure the development of the financial system as well as those that measure the development of the legal and regulatory systems– we observe a negative correlation among them and the discretionary accruals.

b. Multivariate analysis

i. Firm-level variables

Table 5 displays the regressions between the firm-level variables and the earnings management measured according to Model 1 (DAcc1). In all the regressions we observe that the independent variables are jointly statistically significant according to the Wald test. There is not second-order autocorrelation among the variables despite of in some regressions we do observe first-order autocorrelation. This fact does not invalidate our results. Since the very nature of the panel data it is expected

certain level of first-order autocorrelation at the standard 5% confidence level according to Arellano and Bond (1991). Regarding to the moment conditions, both the Sargan and Hansen overidentification tests did not reject the overidentifying restrictions, meaning that the set of instruments are orthogonal to the estimated residuals. Thus, the results reported in Table 5 are robust according to the standard diagnostic tests for the panel data.

In the first regression we observe that the leverage at book value (LevB) impacts negatively the earnings management (DAcc1). This finding indicates that the available resources for the consumption of perks are minimized since higher levels of debt imply larger disbursements of cash flows that otherwise might be used opportunistically by managers. In addition to that, it seems to be that the earnings management is reduced as a consequence of the restrictive debt covenants included in the debt contracts. Creditors and financial institutions will establish more restrictive clauses in the contracts and eventually will demand for higher information about the performance of the firm. As a consequence of a better monitoring, managers will have less room for opportunistic manipulation of the financial statements. This finding allows us to accept our first hypothesis, suggesting that higher leverage impacts negatively on the earnings management.

The results in the second regression suggest that there is a positive and statistically significant relationship between the payout policy (Div1) and the earnings management. In developing countries as in Latin America, managers are reluctant to cut dividends due to its negative consequences and this impacts both upward and downward real earnings management (Anglin et al., 2013). The relative weakness of external corporate governance mechanisms (e.g. institutional systems) implies that shareholders demand for higher dividends. The catalyst of such demand for larger dividends is the discretionary manipulation of financial statements by managers. Agency theory suggests that outside shareholders have a preference for dividends over retained earnings because insiders might squander cash retained within the firm (Easterbrook, 1984; M. C. Jensen, 1986). According to Mitton (2004) this preference for dividends may be even stronger in emerging markets with weak investor protection if shareholders perceive a greater risk of expropriation by insiders in such countries. Our finding show that in order to respond satisfactorily to such preference for dividend, managers take advantage of their discretionary decision power to manipulate opportunistically the financial statements to justify the cash dividend payment. These results allow accepting the hypothesis H2a.⁶

⁶ Similar regressions were computed by using Div2 calculated as the absolute value of the cash dividends divided by the previous year after-tax income. The results were comparable to those shown in Tables 5, 6, 8, 9, and 10 although with some loss of significance in some estimation. For saving space reasons, these results are not included in this work, but they are available upon request to the corresponding author.

The results in the third and fourth regressions show that the ownership concentration (Own1) is negatively related to the earnings management. This finding allows us to accept the hypothesis H3b that states that the higher the ownership concentration the lower the room for opportunistic behavior through earning management. Consequently we can observe that concentrated ownership structures solve efficiently the agency problems by supervising directly the performance of managers. In Latin American countries the ownership structure is highly concentrated as a result of a weak protection of shareholders rights and it seems to be that the controlling power of the majority shareholder prevents the opportunistic behavior of managers.

The last regression in Table 5 is used to test the hypothesis H3a. In this hypothesis we suggested a non-linear relationship between the ownership concentration in hands of the majority shareholder and in hands of executives (measured through the percentage of shares closely held, InsOwn). In this case we used the variable InsOwn and its squared computation (InsOwn²) to test this hypothesis. As it can be seen, there is in fact a quadratic relationship between this variable and the overstatement of the financial reports. As the percentage of closely held shares increases, the earnings management decreases but just up to a certain threshold of ownership concentration. Beyond this level of inside ownership concentration the opportunistic manipulation of the financial statements increases. It seems to be that in the countries of our sample the agency costs and the moral hazard problems are minimized with more concentrated ownership structures. Nevertheless, whenever the controlling shareholder and managers hold more outstanding stocks than those needed for efficient control, the entrenchment and the expropriation agency problems arise. Under this scenario, managers are more inclined to manage the earnings, expropriating in this way certain part of the wealth of the minority shareholders. Such threshold or critical point of ownership concentration might be determined basically by optimizing, for instance, the sixth regression in table 5.⁷ Our findings indicate that when the controlling shareholder and the managers hold no more than 65.02% of the outstanding shares the earnings management is minimized, *ceteris paribus*. When in fact they hold more than this threshold, expropriation and entrenchment agency problems appear which eventually are materialized in the opportunistic manipulation of the financial statements in detriment of the minority shareholders. In turn, this finding allow us to accept the hypothesis H3a which suggested that at low levels of insiders' ownership, alignment of interests exists and the likelihood of earnings management is lower. To make sure that this U-shaped relationship exists, we used the appropriate Lind-

⁷ In order to do so we have to compute the first derivative of this regression relative to the InsOwn variable, and then make it equal to zero. After that we have to solve for InsOwn that represents the point at which the discretionary accruals are minimized. Specifically talking, this solution takes the form: $\frac{\partial \text{DACC1}}{\partial \text{InsOwn}} = -0.0632 + 2 \times 0.0486 \times \text{InsOwn} = 0$ from the sixth regression output in Table 5. Consequently, when InsOwn = 65.02% the earnings management are minimized.

Mehlum test (Lind & Mehlum, 2010). As is can be seen at the bottom of the table in sixth regression, the test allows accepting the U-shaped relationship between these two variables as hypothesized.

In order to avoid problems of under-specification, in the three last regressions the control variables (firm size, profitability, and default risk) are added. We included the firm size (SIZE) because it may affect corporate governance characteristics as well as the level of earnings management (Becker, Defond, Jiambalvo, & Subramanyam, 1998). Our findings show a positive relationship between the firm size and the earnings management. In the same line as Lobo and Zhou (2006), we suggest that larger firms may be more inclined to manage their earnings because the complexity of their operations makes it difficult for users to detect overstatement, and consequently, managers might apply more aggressive accounting policies (Richardson, Tuna, & Wu, 2002). In the same regression we can notice that the profitability of the firm (ROA) is negatively related to earnings management. In other words, we observe that managers from profitable companies do not have the need to manipulate the financial statements.

Finally, in table 5 we observe that the coefficient of the insolvency risk (Z2) variable shows a negative and statistically significant value. By construction, the higher the values of Z2 it means the lower the company's default risk. Consequently, the interpretation of this variable suggests that when the default risk decreases, the discretionary accruals decrease too. And therefore, the lower the bankruptcy risk, the lower the need for manipulating opportunistically the earnings in one or another direction.⁸

ii. Country-level variables

In addition to the internal or firm-level variables studied in the previous section, in table 6 the set of country-level variables derived from the institutional systems are displayed. Recall that these variables are exogenously determined (Aerts, Cheng, & Tarca, 2013). Among these variables we included a set of measures for the development of the financial system (first six regressions) and another set of variables which measure the efficiency of the legal and regulatory systems as corporate governance indicators (last six regressions).

First of all, we observe that the accounting standards are negatively related to the earnings management, indicating that there is less discretionary earnings

⁸ Originally, the models used the Z1 variable. However, for most of the regressions this variable was not statistically significant (not reported here for saving space reasons). Consequently, we decided to use Z2 variable which had much better explicative power than Z1. As it was mentioned above, Z1 is based on the formulation for developed countries (Altman, 1968) whilst Z2 is based on the computation for emerging economies (Altman, 2005). Therefore, we believe that Z2 is more suitable for the goals of this paper than Z1.

management for those countries that adopted international accounting standards (IFRS), which proves the hypothesis H4. Companies using international accounting standards have greater incentives to report transparently because they are subject to higher restrictions. For the period of time considered in the analysis we can state that 35.14% of our observations include firms which reported their statements under the international financial reporting standards (see table 2). And according to the findings shown in table 6, firms reporting under the IFRS reduce the extent of their earnings management in about 1.1%⁹ relative to those firms reporting in local accounting standards.

The first six regressions in table 6 include the country-level variables as measures of the development of the financial system. As stated in our hypothesis, it is likely that the level of financial development heightens the monitoring and scrutiny of accounting figures, because of strengthened laws and regulations for investor protection and by the extension, sophisticated market participants. Enomoto et al. (2014) suggests that the opportunistic behavior of managers is lower under more developed financial systems because higher quality in the accounting information is needed. Our findings reject the hypothesis H6. In fact, earnings management seems to be higher when the banking system is more developed (see the first three regressions in table 6). Contrary to what it was expected, our findings suggest that a more developed banking system increases the discretionary capacity of managers to misreport the financial information. According to this result we can suggest that managers incur in moral hazard problems to overstate the financial statements in order to fulfill the requirements to issue, for instance, more sophisticated financial instruments. This result might be corroborated with the bank concentration index measured by the market share of the three largest banks (Beck et al., 2000), which although it is not reported in this study, it showed a negative and statistically significant relationship with the discretionary accruals measure (DACC1). This finding demonstrate that in less competitive (more monopolistic) banking systems managers have less need for earnings management. Accordingly, it seems to be that, at least concerning the development of the banking system, the earnings management is a reaction to more developed and sophisticated financial instruments and requirements. The variables used to measure this impact are i) the size of the assets of the central bank (CBAGDP) and ii) other financial institutions (OIAGDP) relative to the national GDP, and iii) the size of the private credit by deposit money banks (PCGDP) over the GDP.

Concerning the development of the capital markets –basically corporate bonds and equity– we included the regressions 4 through 6 in table 6. We observe that neither the stock market capitalization nor the private bond market capitalization is statistically significant. Consequently, these two indicators of the development of

⁹ Computed as the arithmetic mean of the coefficient for IFRS variable among the 12 regressions in Table 6.

the capital markets do not determine the earnings management. The international debt issues as a fraction of the GPD is however the only significant variable concerning the development of the capital markets. Nevertheless, its sign is the opposite of the expected one. Once again, we can observe that the exercise of the earnings discretion is a response of higher development of the overall financial system. We suggest that managers in the Latin American countries react before more developed financial systems –more sophisticated financial tools, the response to financial analysts who are tracking the firms’ performance, the disclosure of information to more skilled stakeholders, etc.– by managing opportunistically the earnings. Han et al. (2008) offer an interesting explanation for a relationship like this. They suggest that cultural issues (e.g. uncertainty avoidance and individualism dimensions of national culture) explain managers’ earnings discretion across countries, and that this association varies with the strength of investor protection and the development of the institutional environment. Therefore, we suggest that the widely tested hypothesis in developed economies of an inverse relationship between the development of the financial system and the earnings management needs to be reformulated in emerging markets like those in Latin America, where the protection of the investors’ right are far behind than those observed in more mature (developed) economies.

Despite of the previous results concerning the financial development, the findings about the development of the legal and regulatory systems are overwhelming. For all the six governance indicators (VA, PS, GE, RQ, RL, and CC displayed in regressions 7 through 12 in table 6) about the legal and regulatory systems we observe a negative and statistically significant relationship with the DAcc1 variable. Consequently, we can estate that the better the regulations, the smaller the room for managers to take advantage of their discretionary making power decision on earnings. The legal and regulatory constraints help to protect efficiently the shareholders’ rights, which allow us to accept the formulated H5 hypothesis.

Since we account with a number of variables used as measures for the external governance indicators, and due all these variables are highly correlated (see table 4) we cannot include them altogether in the regression and we opted by introducing these variables individually as displayed in table 6. In order to address this drawback in modeling the discretionary accruals, we decided to apply the principal component factoring technique in order to enter all this variables in the same regression to take advantage of their informative content. The major benefits of this technique are that the factor(s) created is not correlated, on the one hand, and the factor(s) records a large extent of the variability of the individuals variables used in the estimation of the factor(s). Table 7 displays the number of factors generated for the variables used to measure the financial development and the variables used for the legal and regulatory system. In its Panel A we can observe that there are two

factors whose Eigen values are higher than one as the standard discrimination value; which measure the country financial development. These first two factors record about 77.73% of the variability of the six alternative variables used to measure the financial development by country. Panel B however shows that only one factor is enough to record more than 80.00% of the variability of the variables used to measure the legal and regulatory systems.

These three factors have been introduced in the regression analysis shown in Table 8. All the regressions show that the factors are statistically significant. Consequently we can suggest that in a large extent the financial development, as well as the enforcement of the law channeled by the legal and institutional systems, is determinant of the discretionary accruals (either by increasing or decreasing the earnings). Additionally, we have found that the direction of such relationship is in accordance with the results displayed in Table 6: positive for the financial development (Fac1FinDev and Fac2FinDev) and negative for the legal and regulatory systems (Fac1LegEnv). These findings might be considered as robustness checks of our main results.

The last three regressions in table 8 show the models estimated by using the InsOwn variable. These regressions compute the critical values of the ownership in hands of both controlling shareholder and managers at which the discretionary manipulation of the financial statements is minimized. The estimation of such critical values is in the same way as the one explained in the foot note 7. The average value among these three models is 63.03%. If the majority shareholder and the managers hold more than this fraction of the outstanding shares, the likelihood of discretionary accruals increases, which in turn erodes the firms' value. The U-shaped relationship between InsOwn and DAcc1 is tested empirically through the Lind-Mehlum contrast. According to this test, we can accept the fact that there is a statistically significant U-shaped relationship between InsOwn and DAcc1.

iii. Robustness checks

The consistency of our findings is tested by using as the dependent variable the one estimated base on Model 2 (see equation 6). Table 9 shows the main results by using this variable (DAcc2). This variable is slightly different from the DAcc1 because it is adjusted by the change in accounts receivable in order to account for any accounting discretion in collecting the credit sales.

Table 9 includes six regressions accounting for the firm-level variables in addition to the country-level variables. We observe that the leverage (LevB) is negatively correlated with the discretionary accruals. This finding supports what was found above: higher levels of debt reduce the opportunistic behavior of managers

throughout restrictive debt covenants and lower levels of free cash flow available for discretionary use. The payout policy as well as the ownership structure, shows the same relationships as in our previous findings. Therefore, we can observe that our hypotheses concerning the internal corporate governance systems are accepted.

The external corporate governance systems were introduced in the regressions through the different factors which are basically a composite of the financial and legal systems. These three factors (Fac1LegEnv, Fac1FinDev and Fac2FinDev) are statistically significant although some significance power is lost in regressions three and four. Cautiously we might still suggest that besides this lack of significance the results in general are still very consistent by using the alternative variable DAcc2.

iv. Comparative Analysis by Institutional Context

The last part of the empirical analysis offers a comparison by institutional context. In this case, we decided to split the sample in two big groups depending on relative efficiency of their legal and regulatory systems. In order to do so, we computed the average among VA, PS, GE, RQ, RL, and CC by country. Only Chile and Brazil had a positive value and the other countries had a negative average. Consequently, for our period of analysis and sample, Chile and Brazil had relatively better institutional environment than Argentina, Colombia, Mexico, and Peru. Therefore, we re-estimated the regressions taking into consideration these two groups of countries. The results are displayed in Table 10. In this table we observe that the capital structure plays the same role as governance device under a strong (Chile and Brazil) or relatively weak (other countries) institutional context as described above. When we study the dividend policy in details, we observe that the coefficient of Div1 for countries with relatively strong corporate governance (Chile and Brazil in regression 1) is smaller than for countries with relatively weaker governance systems (Argentina, Colombia, Mexico, and Peru in regression 2). This means that the positive impact of the dividend policy on the discretionary accruals is stronger in countries with weak governance systems. We rejected the hypothesis that such coefficients are statistically equal at the standard confidence level of 5%. Consequently, we can accept our hypothesis H2b which suggested higher earnings management under institutional contexts with weak corporate governance systems to achieve certain cash dividend goals than in countries with stronger governance systems. This result might be reinforced with the findings in regressions 3 and 4 of Table 10. Between these couples of regressions we observe that in fact the dividend policy impacts negatively on the accounting discretion for the group compounded by Chile and Brazil, but positive for all the other countries with relatively weaker institutional systems.

Concerning the corporate ownership structure measured through the Own1 variable we observe that it is an efficient corporate governance tool in countries with better protection of the investors' rights (e.g. Chile and Brazil), impacting negatively on the discretionary capacity of managers to manage earnings. However, under institutional contexts with weak governance indicators (those included in the second regression such as Argentina, Colombia, Mexico, and Peru) more concentrated ownership structures lead to higher managerial discretion. Similar findings are shown when InsOwn is considered (see regressions 3 and 4 in Table 10). For instance, in the third regression we still observe that the U-shaped relationship between InsOwn and DAcc1 holds in the institutional contexts with relatively better corporate governance systems. However, such relationship is the opposite in institutional environment with relatively poor protection of investors' rights (see regression 4). Where we observe that in fact, for most of the range of the InsOwn variable (96.56%) the earnings management increases as the InsOwn increases too. Moreover, as it can be seen at the bottom of the fourth regression, according to the Lind-Mehlum test, it does not exist a U-shaped relationship between InsOwn and DAcc1.

Finally, we still observe that independently of how good the corporate governance is under the two institutional environments, the adoption of the International Financial Reporting Standards (IFRS) still reduce the managerial discretionary accounting.

5.CONCLUSIONS

In this paper we analyzed the impact of internal and external variables on the managers' discretionary behavior for a sample of companies from Argentina, Brazil, Chile, Colombia, Mexico, and Peru.

It is a fact in terms of corporate governance that accounting discretion is determined by factors that are internal to companies, but also by the characteristics of the institutional environment in the Latin American context.

The varying efficiency of these governance mechanisms is reflected in the way in which accounting discretion is performed. An example of this is the dividend policy, which states that in order to meet the demand of dividends by shareholders; managers are motivated to manipulate accounting information in order to meet the required dividends.

Additionally, we concluded that the ownership structure is a double hedge sword as a governance device. The way the corporate ownership is shaped might work and an efficient governance tool, but also it might foster moral hazard problems.

In relation to the country-level variables we found that those associated with the development of the financial system behaved oppositely to the expected. Therefore we conclude that the financial systems are in a stage of such premature development, that managers make use of accounting discretion to manipulate financial information in order to be active in these markets. In immature markets, with large information gaps and opacity investors may not discriminate between what companies give good or bad information (Akerlof, 1970; Paolo Saona & Valledado, 2012). Consequently, before inefficient financial markets in Latin America, managers have more room to manipulate the financial statements. Regarding the legal and regulatory system, we conclude that it is an efficient corporate governance mechanism, since the opportunistic behavior of managers is reduced.

We derive out of this that is necessary for governments, institutions, and policy makers to develop policies that promote market efficiency. Similarly, policies that protect more effectively the rights of minority shareholders, because they have a greater risk of expropriation in emerging markets than in developed markets. Consequently, we argue that more efficient markets might eliminate opacity and information asymmetries, which eventually will decrease managers' discretionary power.

Finally, as a future extension of this study we suggest the analysis of the impact of the board of directors in the accounting discretion of managers in emerging countries.

Table 1. Panel Data Composition

Country	Observations	Firms	Avg. Obs. per Country
Argentina	231	35	6.60
Brazil	1,686	270	6.24
Chile	797	142	5.61
Colombia	186	33	5.64
Mexico	917	112	8.19
Peru	733	120	6.11
Total	4,550	712	6.39

Table 2: Descriptive Statistics by Variables

Variable	Acronym	Definition	Obs	Mean
Discretionary accruals	DAcc1	Discretionary accruals based on Model 1	4545	0.0203
	DAcc12	Discretionary accruals based on Model 2	4545	0.0697
Capital structure	LevB	Leverage at book values	4550	0.4865
	LevM	Leverage at market values	4475	0.1742
Dividend policy	Div1	Payout ratio: DPS/EPS	4550	1.0163
	Div2	Payout ratio: Cash Dividend / NI(t-1)	4381	0.0231
Ownership structure	Own1	Ownership structure: % shares in hands of the majority shareholder	4546	0.2713
	InsOwn	Ownership closely held: % shares in hands of managers and majority shareholder	2556	0.5871
Firm size	Size	Ln(TA)	4550	6.6236
Profitability	ROA	NI/TA	4550	0.0688
Liquidity risk	Z1	Altman Z-Score for developed economies	4550	4.7765
	Z2	Altman Z-Score for emerging markets	4550	7.5574
Financial Devlpmnt.	CBAGDP	Central bank assets / GDP	4549	6.5495
	OIAGDP	Other financial institutions assets /GDP	4549	10.7839
	PCGDP	Private credit by deposit money banks / GDP	4549	27.3127
	SMKGDP	Stock market capitalization / GDP	4549	43.7889
	IDGDP	International debt issues / GDP	4549	9.2647
	PBGDP	Private bond market capitalization / GDP	4549	11.1535
Legal System	IFRS	0 if local and 1 if IFRS	4550	0.3514
	VA	Voice & accountability	4549	0.3852
	PS	Political stability	4548	-0.2409
	GE	Governance effectiveness	4549	0.1678
	RQ	Regulatory quality	4549	0.4132
	RL	Rule of law	4549	-0.1236
	CC	Control of corruption	4549	0.0990

Table 3. Descriptive statistics of the alternative measures of earnings management for the whole sample and by country

Panel A: Earnings Management for the whole sample					
Variable	Obs.	Mean	Std. Error	Std. Dev.	p-value
DAcc1	4545	0.0203	0.0003	0.0229	(0.0000)
DAcc2	4545	0.0697	0.0013	0.0882	(0.0000)
Panel B: Earnings Management (Model 1) by country					
DAcc11	Obs.	Mean	Std. Error	Std. Dev.	p-value
Argentina	231	0.0204	0.0012	0.0188	(0.0000)
Brazil	1788	0.0177	0.0004	0.0184	(0.0000)
Chile	789	0.0238	0.0012	0.0323	(0.0000)
Colombia	87	0.0216	0.0042	0.0394	(0.0000)
Mexico	917	0.0203	0.0006	0.0196	(0.0000)
Peru	733	0.0227	0.0008	0.0225	(0.0000)
Panel C: Earnings Management (Model 2) by country					
DAcc2	Obs.	Mean	Std. Error	Std. Dev.	p-value
Argentina	231	0.0670	0.0051	0.0770	(0.0000)
Brazil	1788	0.0730	0.0021	0.0905	(0.0000)
Chile	789	0.0708	0.0034	0.0962	(0.0000)
Colombia	87	0.0595	0.0071	0.0658	(0.0000)
Mexico	917	0.0616	0.0027	0.0829	(0.0000)
Peru	733	0.0726	0.0031	0.0852	(0.0000)

Table 4: Correlation Matrix

	Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	DAcc1	1.000																							
2	DAcc2	0.928	1.000																						
3	LevB	-0.226	-0.153	1.000																					
4	LevM	-0.126	-0.091	0.330	1.000																				
5	Div1	-0.006	-0.001	0.011	-0.024	1.000																			
6	Div2	0.221	0.234	-0.046	-0.064	0.068	1.000																		
7	Own1	0.043	0.037	-0.061	-0.032	-0.014	0.031	1.000																	
8	InsOwn	0.072	0.031	-0.040	0.008	-0.032	0.061	0.408	1.000																
9	Size	-0.106	-0.092	0.470	0.145	0.007	0.016	-0.043	-0.122	1.000															
10	ROA	0.014	0.024	-0.314	-0.193	0.123	0.132	0.001	0.044	-0.233	1.000														
11	Z1	0.004	0.018	-0.687	-0.210	0.010	0.010	0.070	0.025	-0.371	0.401	1.000													
12	Z2	0.075	0.059	-0.375	-0.234	0.051	0.079	0.161	0.084	-0.302	0.069	0.439	1.000												
13	IFRS	-0.099	-0.083	0.022	0.067	-0.012	-0.084	0.128	-0.091	0.071	0.089	-0.023	-0.188	1.000											
14	CBAGDP	-0.121	-0.076	0.199	0.351	-0.014	-0.024	-0.212	-0.210	0.242	0.009	-0.149	-0.346	0.149	1.000										
15	OIAGDP	-0.084	-0.048	0.049	-0.004	-0.082	-0.054	0.104	0.008	0.169	-0.052	-0.050	-0.039	0.389	0.185	1.000									
16	PCGDP	-0.126	-0.084	0.158	0.273	-0.064	-0.079	-0.060	-0.126	0.238	-0.005	-0.136	-0.292	0.509	0.668	0.750	1.000								
17	SMKGDP	-0.080	-0.052	0.117	0.153	-0.083	-0.064	0.042	0.002	0.156	0.028	-0.100	-0.246	0.513	0.343	0.809	0.862	1.000							
18	IDGDP	-0.057	-0.063	0.059	0.208	-0.051	-0.126	-0.152	-0.022	0.051	0.026	-0.068	-0.374	0.033	0.275	-0.047	0.187	0.230	1.000						
19	PBGDP	-0.157	-0.106	0.179	0.253	-0.052	-0.095	-0.102	-0.164	0.318	-0.002	-0.138	-0.405	0.368	0.735	0.648	0.836	0.644	0.171	1.000					
20	va	0.077	0.093	-0.038	-0.182	0.011	0.162	0.154	0.072	-0.080	-0.052	0.048	0.316	-0.083	-0.051	0.221	-0.047	-0.014	-0.552	-0.098	1.000				
21	ps	0.061	0.074	-0.014	-0.095	0.005	0.143	0.065	0.028	-0.050	-0.075	0.006	0.182	-0.150	0.097	0.063	-0.083	-0.174	-0.404	-0.092	0.869	1.000			
22	ge	0.100	0.090	-0.158	-0.344	0.019	0.132	0.197	0.128	-0.157	-0.087	0.122	0.460	-0.170	-0.542	0.123	-0.381	-0.256	-0.598	-0.430	0.769	0.669	1.000		
23	rq	0.111	0.096	-0.151	-0.292	0.028	0.137	0.190	0.120	-0.168	-0.050	0.114	0.448	-0.041	-0.587	0.083	-0.315	-0.174	-0.699	-0.438	0.657	0.507	0.906	1.000	
24	rl	0.091	0.091	-0.110	-0.255	0.029	0.157	0.172	0.075	-0.129	-0.077	0.090	0.455	-0.042	-0.294	0.181	-0.168	-0.135	-0.689	-0.285	0.892	0.781	0.919	0.871	1.000
25	cc	0.108	0.102	-0.115	-0.274	0.031	0.162	0.163	0.100	-0.160	-0.068	0.097	0.457	-0.105	-0.350	0.122	-0.224	-0.151	-0.671	-0.368	0.873	0.752	0.924	0.890	0.982

Table 5. Multivariate analysis: Dependent variable is DAcc1 and the independent variables are the internal corporate governance determinants

The table shows coefficient estimates. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0223*** (0.00166)	0.0278*** (0.000309)	0.0290*** (0.000258)	0.0468*** (0.000490)	0.0355*** (1.42e-06)	0.0256*** (5.61e-06)
LevB	-0.00602* (0.00322)	-0.0168*** (0.000257)	-0.0146*** (0.000135)	-0.0301*** (0.000433)	-0.0277*** (2.43e-06)	-0.0256*** (3.29e-06)
Div1		0.000165*** (2.19e-06)	1.64e-07 (5.88e-07)	2.15e-05*** (7.24e-07)	2.21e-05*** (3.42e-09)	2.63e-05*** (5.71e-09)
Own1			-0.00854*** (0.0001)	-0.00864*** (5.85e-05)		
InsOwn					0.0148 (5.28e-07)	-0.0632*** (1.47e-05)
InsOwn ²						0.0486*** (1.64e-05)
<i>Critical Value</i>						0.6502
SIZE				0.000296*** (6.07e-05)	0.000324*** (4.83e-07)	0.000335*** (9.24e-07)
ROA				-0.00815*** (0.000550)	-0.0228*** (5.07e-06)	-0.0195*** (6.43e-06)
Z1				-0.00113*** (1.97e-05)	-0.00150*** (2.19e-07)	-0.00143*** (1.65e-07)
Obs.	11,072	6,449	4,541	4,541	2,940	2,940
No. Ident.	949	841	710	710	624	624
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes
Temporal Effect	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
No. Instr.	120	275	379	379	426	426
Wald-test	3.50	47.05	22518	29186	41101	905.00
p-value	0.061	0.000	0.000	0.000	0.000	0.000
AR(1)	-6.06	-4.84	-3.418	-3.57	-2.295	-2.295
p-value	0.000	0.000	0.001	0.020	0.000	0.022
AR(2)	-2.15	-1.15	1.162	0.866	1.232	1.232
p-value	0.320	0.249	0.245	0.386	0.255	0.218
Sargan-Hansen-test	31.48	37.61	40.60	40.12	40.30	45.22
p-value	0.207	0.165	0.112	0.128	0.218	0.736
Lind Mehlum-test	-	-	-	-	-	1876.73
p-value	-	-	-	-	-	0.000

Table 6. Multivariate analysis: Dependent variable is DAcc1 and the independent variables are the internal and external corporate governance determinants

The table shows coefficient estimates. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

VARIABLES	Development of the Financial System						Development of the Legal and Regulatory System					
	Banking System			Capital Markets			(7)	(8)	(9)	(10)	(11)	(12)
	(1)	(2)	(3)	(4)	(5)	(6)						
Constant	0.0379*** (0.000623)	0.0390*** (0.000590)	0.0382*** (0.000630)	0.0383*** (0.000607)	0.0360*** (0.000597)	0.0380*** (0.000636)	0.0354*** (0.000609)	0.0379*** (0.000634)	0.0375*** (0.000625)	0.0366*** (0.000647)	0.0376*** (0.000645)	0.0364*** (0.000697)
LevB	-0.0326*** (0.000486)	-0.0326*** (0.000491)	-0.0327*** (0.000487)	-0.0327*** (0.000488)	-0.0324*** (0.000504)	-0.0330*** (0.000484)	-0.0325*** (0.000431)	-0.0332*** (0.000348)	-0.0326*** (0.000488)	-0.0333*** (0.000485)	-0.0329*** (0.000484)	-0.0331*** (0.000466)
Div1	5.26e-05*** (5.30e-07)	5.33e-05*** (5.98e-07)	5.29e-05*** (5.52e-07)	5.23e-05*** (5.09e-07)	5.64e-05*** (5.62e-07)	5.16e-05*** (5.61e-07)	4.80e-05*** (4.19e-07)	5.33e-05*** (5.56e-07)	5.20e-05*** (6.77e-07)	4.98e-05*** (6.02e-07)	4.90e-05*** (5.84e-07)	4.99e-05*** (5.00e-07)
Own1	-0.00460*** (0.00006)	-0.00533*** (0.00005)	-0.00479*** (0.00006)	-0.00479*** (0.00001)	-0.00351*** (0.00001)	-0.00468*** (0.00001)	-0.00604*** (0.00001)	-0.00479*** (0.00001)	-0.00516*** (0.00001)	-0.00528*** (0.00001)	-0.00531*** (0.00001)	-0.00554*** (0.00001)
SIZE	0.00158*** (6.70e-05)	0.00147*** (6.65e-05)	0.00158*** (6.77e-05)	0.00162*** (6.75e-05)	0.00168*** (6.95e-05)	0.00170*** (6.84e-05)	0.00178*** (7.18e-05)	0.00185*** (7.11e-05)	0.00169*** (6.70e-05)	0.00179*** (6.73e-05)	0.00179*** (7.25e-05)	0.00194*** (8.33e-05)
ROA	-0.0116*** (0.000605)	-0.0119*** (0.000607)	-0.0113*** (0.000628)	-0.0112*** (0.000602)	-0.0121*** (0.000617)	-0.0105*** (0.000626)	-0.0126*** (0.000798)	-0.00846*** (0.000754)	-0.0105*** (0.000609)	-0.00988*** (0.000683)	-0.0106*** (0.000624)	-0.0108*** (0.000698)
Z2	-0.000941*** (2.36e-05)	-0.000952*** (2.27e-05)	-0.000942*** (2.46e-05)	-0.000942*** (2.34e-05)	-0.000928*** (2.54e-05)	-0.000943*** (2.38e-05)	-0.000940*** (2.23e-05)	-0.000920*** (2.11e-05)	-0.000936*** (2.42e-05)	-0.000941*** (2.42e-05)	-0.000936*** (2.28e-05)	-0.000943*** (2.37e-05)
IFRS	-0.0109*** (0.000245)	-0.0115*** (0.000288)	-0.0110*** (0.000285)	-0.0106*** (0.000320)	-0.0108*** (0.000236)	-0.0106*** (0.000285)	-0.0108*** (0.000252)	-0.0108*** (0.000261)	-0.0108*** (0.000237)	-0.0109*** (0.000255)	-0.0111*** (0.000238)	-0.0110*** (0.000244)
CBAGDP	9.41e-05*** (2.85e-05)											
OIAGDP		6.80e-05*** (1.00e-05)										
PCGDP			1.80e-05** (7.44e-06)									
SMKGDP				-8.05e-07 (3.92e-06)								
IDGDP					0.000156*** (1.41e-05)							
PBGDP						-1.86e-05 (2.38e-05)						
VA							-0.00581*** (0.000149)					
PS								-0.00500*** (7.37e-05)				
GE									-0.00198*** (0.000349)			

RQ											-0.00219*** (0.000229)		
RL												-0.00222*** (0.000289)	
CC													-0.00250*** (0.000193)
Obs.	4,540	4,540	4,540	4,540	4,540	4,540	4,540	4,539	4,540	4,540	4,540	4,540	4,540
No. Ident.	710	710	710	710	710	710	710	710	710	710	710	710	710
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Temporal Effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. Instr.	380	379	379	379	379	379	377	377	377	377	377	377	377
Wald-test	49691.84	41832.43	28749.40	26110.35	17565.75	28749.40	26739.91	25453.20	27655.25	31240.01	29485.08	50642.59	
p-value	0.000	0.000	0.000	0.000	0.0000	0.000	0.000	0.000	0.000	0.000	0.0000	0.000	
AR(1)	-3.56	-3.55	-3.54	-3.45	-3.40	-3.54	-0.354	-3.55	-3.56	-3.61	-3.56	-3.54	
p-value	0.303	0.000	0.000	0.001	0.000	0.002	0.000	0.000	0.000	0.000	0.005	0.000	
AR(2)	1.07	1.05	0.83	1.16	1.19	0.83	0.83	0.75	0.81	1.02	0.83	0.79	
p-value	0.000	0.292	0.303	0.246	0.494	0.404	0.494	0.456	0.421	0.306	0.406	0.428	
Sargan-Hansen-test	378.50	378.50	382.13	381.96	378.50	382.13	378.62	382.67	378.50	36.23	379.72	380.94	
p-value	0.470	0.297	0.337	0.336	0.355	0.334	0.340	0.288	0.316	0.472	0.326	0.310	

This table continues from the previous page.

Table 7. Principal component factoring analysis. Financial development and legal and regulatory systems.

Variables	Factor	Eigenvalue	Difference	Proportion	Cumulative
Panel A: Financial Development Variables					
CBAGDP	Factor1	3.5559	2.4479	0.5927	0.5927
OIAGDP	Factor2	1.1080	0.2946	0.1847	0.7773
PCGDP	Factor3	0.8133	0.4773	0.1356	0.9129
SMKGDP	Factor4	0.3360	0.2174	0.056	0.9689
IDGDP	Factor5	0.1186	0.0505	0.0198	0.9886
PBGDP	Factor6	0.0681	0.0000	0.0114	1.0000
Panel B: Legal and Regulatory Systems					
VA	Factor1	5.0000	4.3130	0.8333	0.8333
PS	Factor2	0.6870	0.5310	0.1145	0.9478
GE	Factor3	0.1561	0.0701	0.0260	0.9738
RQ	Factor4	0.0859	0.0346	0.0143	0.9882
RL	Factor5	0.0514	0.0318	0.0086	0.9967
CC	Factor6	0.0196	0.0000	0.0033	1.0000

Table 8. Regression Results including Factors for Financial Development and Legal and Regulatory Systems. Dependent variable DAcc1

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0371*** (0.000637)	0.0382*** (0.000609)	0.0368*** (0.000642)	0.00305*** (6.12e-05)	0.00290*** (1.97e-05)	0.00169*** (3.07e-05)
LevB	-0.0330*** (0.000452)	-0.0323*** (0.000504)	-0.0322*** (0.000480)	-0.00999*** (5.63e-06)	-0.00757*** (3.36e-06)	-0.00858*** (6.59e-06)
Own1	-0.00570*** (5.87e-05)	-0.00395*** (5.45e-05)	-0.00453*** (6.12e-05)			
InsOwn				-0.0485*** (2.51e-05)	-0.0479*** (5.35e-06)	-0.0479*** (9.23e-06)
InsOwn ²				0.0381*** (2.52e-05)	0.0380*** (6.73e-06)	0.0377*** (9.68e-06)
<i>Critical Value</i>				<i>0.63648</i>	<i>0.63026</i>	<i>0.63528</i>
Div1	4.94e-05*** (4.89e-07)	5.51e-05*** (5.74e-07)	5.53e-05*** (6.45e-07)	7.47e-06* (1.47e-08)	2.34e-05 (1.72e-08)	1.71e-05* (1.06e-08)
SIZE	0.00187*** (7.07e-05)	0.00158*** (6.91e-05)	0.00188*** (7.41e-05)	0.00223*** (1.71e-06)	0.00204*** (1.67e-06)	0.00226*** (1.86e-06)
ROA	-0.0101*** (0.000736)	-0.0126*** (0.000632)	-0.0131*** (0.000820)	-0.0411*** (4.11e-06)	-0.0361*** (7.63e-06)	-0.0369*** (5.16e-06)
Z1	-0.000942*** (2.33e-05)	-0.000927*** (2.56e-05)	-0.000912*** (2.65e-05)	-0.00136*** (8.47e-07)	-0.00139*** (3.90e-07)	-0.00145* (9.91e-07)
IFRS	-0.0111*** (0.000259)	-0.0111*** (0.000310)	-0.0120*** (0.000335)	-0.00851*** (4.59e-06)	-0.00691*** (2.11e-06)	-0.00711*** (2.87e-06)
Fac1LegEnv	-0.00197** (8.79e-05)		-0.00277*** (0.000100)	-0.00228** (8.12e-07)		-0.000876*** (5.93e-07)
Fac1FinDev		0.000437*** (0.000146)	0.000963*** (0.000144)		0.00146*** (1.18e-05)	0.00126*** (1.56e-05)
Fac2FinDev		0.000653*** (9.02e-05)	0.00131*** (0.000108)		0.00194*** (3.90e-06)	0.00149*** (4.43e-06)
Obs.	4,539	4,540	4,539	2,892	2,939	2,892
No. Ident.	710	710	710	622	624	622
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes
Temporal Effect	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
No. Instr.	407	407	407	407	407	369
Wald-test	34805.22	55057.05	2765.39	39551.62	17705.06	13551.2
p-value	0.00	0.215	0.000	0.000	0.000	0.000
AR(1)	-2.185	-3.54	-3.55	-3.55	-2.62	-2.18
p-value	0.029	0.0289	0.000	0.000	0.000	0.013
AR(2)	1.241	0.81	0.84	0.83	1.241	-1.55
p-value	0.215	0.416	0.389	0.405	0.215	0.121
Sargan-Hansen-test	378.60	381.49	382.15	382.09	337.09	265.76
p-value	0.727	0.303	0.295	0.289	0.858	0.727
Lind Mehlum-test	-	-	-	885.67	943.98	704.56
p-value	-	-	-	0.000	0.000	0.000

Table 9. Robustness Analysis including Factors for Financial Development and Legal and Regulatory Systems. Dependent variable DAcc2

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.00851*** (0.00240)	0.0261*** (0.00257)	0.0251*** (0.00276)	-0.00666*** (4.12e-05)	0.0209*** (0.000128)	0.0231*** (0.000120)
LevB	-0.0471*** (0.00232)	-0.0495*** (0.00232)	-0.0491*** (0.00237)	-0.0555** (4.61e-05)	-0.0536** (2.41e-05)	-0.0539** (4.84e-05)
Div1	9.07e-05 (1.06e-05)	6.40e-05** (9.03e-06)	6.81e-05* (1.01e-05)	6.46e-05*** (1.23e-07)	4.19e-05*** (1.33e-07)	1.51e-05*** (1.44e-07)
Own1	-0.00602*** (0.000642)	-0.00119* (0.000625)	-0.00125** (0.000633)			
InsOwn				-0.05078*** (0.000115)	-0.04169*** (0.000113)	-0.04321*** (0.000142)
InsOwn ²				0.03700*** (0.000115)	0.02993*** (0.000112)	0.03064*** (0.000146)
<i>Critical value</i>				<i>0.06862</i>	<i>0.06965</i>	<i>0.070495</i>
IFRS	-0.0264*** (0.00121)	-0.0369*** (0.00128)	-0.0372*** (0.00131)	-0.0293*** (2.57e-05)	-0.0516*** (2.44e-05)	-0.0511*** (2.73e-05)
SIZE	0.00152*** (0.000478)	-0.000359 (0.000494)	-0.000151 (0.000534)	0.00349*** (7.38e-06)	0.00133*** (5.80e-06)	0.000993*** (1.27e-05)
ROA	-0.279*** (0.00370)	-0.261*** (0.00383)	-0.260*** (0.00405)	-0.263 (2.99e-05)	-0.250 (4.74e-05)	-0.244* (4.45e-05)
Z1	-0.00183 (5.69e-05)	-0.00184* (6.53e-05)	-0.00184* (6.57e-05)	-0.00248* (1.64e-06)	-0.00221 (1.37e-06)	-0.00219* (7.80e-07)
Fac1LegEnv	-0.00198*** (0.000494)		0.000660 (0.000506)	-0.00774*** (9.80e-06)		-0.00434*** (1.04e-05)
Fac1FinDev		0.0114*** (0.000761)	0.0116*** (0.000768)		0.0177*** (8.33e-05)	0.0168*** (0.000120)
Fac2FinDev		0.000607 (0.000593)	0.000770 (0.000603)		0.000157*** (1.41e-05)	0.00166*** (1.59e-05)
Obs.	4,539	4,540	4,539	2,892	2,939	2,892
No. Ident.	710	710	710	622	624	622
Industry Effect	Yes	Yes	Yes	Yes	Yes	Yes
Temporal Effect	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
No. Instr.	407	407	407	407	407	407
Wald-test	48807.25	55057.05	53761.48	34485.02	27941.44	60452.98
p-value	0.000	0.000	0.000	0.000	0.872	0.000
AR(1)	-7.18	-354	-3.55	-3.56	-3.57	-3.57
p-value	0.013	0.000	0.000	0.000	0.000	0.000
AR(2)	-0.477	0.81	0.77	0.72	0.77	0.70
p-value	0.634	0.416	0.440	0.634	0.522	0.485
Sargan-Hansen-test	363.2	381.49	378.79	383.50	379.79	378.74
p-value	0.872	0.303	0.324	0.266	0.324	0.287
Lind Mehlum-test	-	-	-	432.36	607.98	350.83
p-value	-	-	-	0.000	0.000	0.000

Table 10. Comparative Analysis by Institutional Context. Dependent variable DAcc1

VARIABLES	(1)	(2)	(3)	(4)
	Chile+Brazil	Other Countries	Chile+Brazil	Other Countries
Constant	0.0298*** (6.13e-06)	0.0423*** (4.67e-06)	0.00834*** (3.62e-05)	0.0448*** (0.000630)
LevB	-0.0141*** (2.65e-06)	-0.0301*** (1.90e-06)	-0.00611** (2.30e-05)	-0.0311* (0.000352)
Div1	0.000026*** (1.39e-08)	0.000043*** (9.98e-09)	-0.000100*** (7.18e-08)	0.000019** (6.97e-07)
Own1	-0.00620*** (3.09e-07)	0.00167*** (1.25e-06)		
InsOwn			-0.0385*** (5.65e-05)	0.0197*** (0.000319)
InsOwn ²			0.0303*** (4.33e-05)	-0.0102*** (0.000327)
<i>Critical Value</i>			0.6353	0.9656
IFRS	-0.0120*** (6.28e-06)	-0.00388*** (7.32e-07)	-0.00848* (1.70e-05)	-0.000287** (5.68e-05)
SIZE	0.00155*** (3.10e-06)	-0.000344*** (8.17e-07)	0.00199*** (2.53e-06)	-0.00220*** (7.43e-05)
ROA	-0.0314*** (3.66e-06)	0.0175*** (3.08e-06)	0.0241 (2.39e-05)	0.0199* (0.000456)
Z1	-0.00114*** (4.96e-07)	-0.000914*** (4.07e-08)	-0.00127*** (8.48e-07)	-0.00122*** (8.43e-06)
Fac1LegEnv	0.00575*** (3.21e-06)	-0.00115*** (4.45e-07)	0.00283*** (6.02e-06)	0.00587*** (6.02e-05)
Fac1FinDev	0.000663*** (1.86e-06)	-0.00230*** (1.27e-06)	0.000643* (8.96e-06)	0.0199** (0.000169)
Fac2FinDev	0.00287*** (2.67e-06)	0.00122*** (5.85e-07)	0.00127* (6.22e-06)	-0.00147** (3.55e-05)
Obs.	2,573	1,966	1,992	900
No. Ident.	411	299	376	246
Industry Effect	Yes	Yes	Yes	Yes
Temporal Effect	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
No. Instr.	377	369	324	324
Wald-test	17704.22	14405.10	15533.88	14255.25
p-value	0.000	0.000	0.000	0.000
AR(1)	-2.625	-2.47	-3.793	-3.44
p-value	0.009	0.223	0.000	0.000
AR(2)	1.217	-1.55	-1.60	-1.47
p-value	0.223	0.121	0.563	0.379
Sargan-Hansen-test	337.1	265.76	170.80	212.50
p-value	0.858	0.867	0.889	0.485
Lind Mehlum-test	-	-	390.16	8.310
p-value	-	-	0.000	0.744

Appendix

Leverage:

$$i) LevB = \frac{TD_{it}}{TA_{it}}$$

Where TD_{it} is the total liabilities at the year t for the i firm, and TA_{it} is the total assets.

$$ii) LevM = \frac{TD_{it}}{TD_{it} + MkCptz_{it}}$$

Where $MkCptz_{it}$ is the market capitalization computed as the product among the year-end close price per share P_{it} and the number of shares outstanding per i firm.

Dividends:

$$i) Div1_{it} = \left| \frac{DPS_{it}}{EPS_{it}} \right|$$

Where DPS_{it} is the cash dividend per share and EPS_{it} corresponds to the earnings per share.

$$ii) Div2_{it} = \left| \frac{Dividends_{it}}{NI_{it-1}} \right|$$

Where $Dividends_{it}$ are the cash annual dividends and NI_{it-1} is the net income.¹⁰

Ownership structure:

$$i) Own1_{it}$$

Which corresponds to the shares held by the controlling (majority) shareholder.

$$ii) PctShClHeld_{it}$$

Which represents the shares held by insiders –majority/controlling shareholder plus the ownership of executives–.

Firm size:

$$i) Size_{it} = \ln(TA_{it})$$

The natural logarithm transformation is the usual method when dealing with a variable which takes large and positive values.

Profitability:

$$i) ROA_{it} = \frac{EBT_{it}}{TA_{it}}$$

Where EBT_{it} are the earnings before taxes.

Default risk:

$$i) Z1 = 1.2WC_{it} + 1.4RE_{it} + 3.3EBIT_{it} + 0.6MvE_{it} + 0.99S_{it}$$

Where WC_{it} is the working capital/total assets; RE_{it} is the retained earnings/total assets; $EBIT_{it}$ the earnings before interest and tax/total assets; MvE_{it} is the market value of equity/total liabilities; and S_{it} is the sales/total assets.

$$ii) Z2 = 6.56WC_{it} + 3.26RE_{it} + 6.72EBIT_{it} + 1.05BvE_{it} + 3.25$$

Where BvE_{it} is the book value of equity/total liabilities.

Accounting Standards:

$$i) Accounting\ Standards = \begin{cases} 1, & \text{if the country uses IFRS} \\ 0, & \text{if the country does not use IFRS} \end{cases}$$

Legal and Regulatory Systems:

All the following legal system variables were obtained from Kaufmann et al. (2011). The indexes range from approximately -2.5 (weak) to 2.5 (strong) governance performance

i) **VA.** Voice and accountability.

ii) **PS.** Political stability and absence of violence/terrorism

iii) **GE.** Government effectiveness

¹⁰ We used this widely used measure for dividend payout ratio even though many times the paid cash dividends may come from earnings attained in different years.

- iv) **RQ.** Regulatory quality
- v) **RL.** Rule of law
- vi) **CC.** Control of corruption

Financial Development:

All the following financial development variables were obtained from Beck et al. (2000).

- i) **CBAGDP** is the claims on domestic real nonfinancial sector by the Central Bank as a share of GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t + F_{t-1}}{P_{et} + P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$, where F is Central Bank claims, P_e is end-of period Consumer Price Index (CPI), and P_a is average annual CPI.
- ii) **OIAGDP** is the claims on domestic real nonfinancial sector by other financial institutions as a share of GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t + F_{t-1}}{P_{et} + P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$, where F is other financial institutions' claims, P_e is end-of period Consumer Price Index (CPI), and P_a is average annual CPI.
- iii) **PCGDP** is the private credit by deposit money banks to GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t + F_{t-1}}{P_{et} + P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$; where F is the credit to the private sector, P_e is end-of period CPI, and P_a is average annual CPI.
- iv) **SMKGDP** which is the value of listed shares to GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t + F_{t-1}}{P_{et} + P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$; where F is the stock market capitalization, P_e is end-of period CPI, and P_a is average annual CPI.
- v) **IDGDP** computed as the International Debt Securities (amount outstanding) as a share of GDP.
- vi) **PBGDP** corresponds to the private domestic debt securities issued by financial institutions and corporations as a share of GDP, calculated using the following deflation method: $\frac{0.5 \left[\frac{F_t + F_{t-1}}{P_{et} + P_{et-1}} \right]}{\left[\frac{GDP_t}{P_{at}} \right]}$; where F is the amount outstanding of private domestic debt securities, P_e is end-of period CPI, and P_a is average annual CPI.

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